Khulna University of Engineering & Technology
B.Sc. Engineering Special Backlog Examination, 2020
Department of Materials Science and Engineering

Math 1227
(Differential Equations and Transform Analysis)

Time: 3 hours
Full Marks: 210

N.B. (i) Answer **ANY THREE** questions from each section in separate scripts.
(ii) Figure in the right margin indicates full marks

Section A

(Answer **ANY THREE** questions from this section in **answer script A**)

1. (a) Obtain the differential equation of which $y = A\cos\alpha x + B\sin\alpha x$ is a solution, where A and B are arbitrary constants and α is a fixed constant. (11)

   (b) Solve $xdy - ydx = \sqrt{(x^2 + y^2)}dx$. (12)

   (c) Solve $y\sin 2x dx - (y^2 + \cos^2 x)dy = 0$. (12)

2. (a) Solve $(D^2 + D)y = x\cos x;\ \ D = \frac{d}{dx}$. (11)

   (b) Solve $(D^2 - 9D + 20)y = 20x;\ D = \frac{d}{dx}$. (12)

   (c) Use the method of variation of parameter to solve $\frac{d^2y}{dx^2} + y = \sec x$. (12)

3. (a) Solve in series, $(1 - x^2)y'' - 2xy' + 3y = 0$. (20)

   (b) Prove that $P_n(x) = \frac{1}{2^n n!}\frac{d^n}{dx^n}(x^2 - 1)^n$. Also, find $P_3(x)$. (15)

4. (a) Write Bessel's differential equation of zero order. Prove that $J_{-n}(x) = (-1)^n J_n(x)$, where $J_n(x)$ is the Bessel's function of first kind and n is a positive integer. (12)

   (b) Find the value of $J_2'(x)$ in terms of $J_0(x)$ and $J_1(x)$, where $J_n(x)$ is the Bessel's function of final kind. (11)

   (c) Prove that (i) $\int_{-1}^{1} P_n(x)dx = 2,\ \ if\ n = 0$
   (ii) $\int_{-1}^{1} P_n(x) = 0,\ \ \ if\ n \geq 1$. (12)

Section B

(Answer **ANY THREE** questions from this section in **answer script B**)

5. (a) Find the inverse Laplace transform of $\frac{2s^2-4}{(s+1)(s-2)(s-3)}$ using the Heaviside expansion formula. (10)

(b) Find the series of sine and cosine of multiples of x which represents $f(x)$ in $(-\pi,\pi)$, where $f(x)=0$, when $-\pi < x < 0$ and $\frac{\pi x}{4}$, when $0 < x < \pi$ and hence prove that (15)

$$1 - {}^{1}/_{3} + {}^{1}/_{5} - \cdots = \frac{\pi}{4}.$$

(c) Expand the function $f(x) = x,\ 0 < x < 2$ in a half-range Fourier sine series. (10)

6. (a) Expand f(x) in Fourier series, where $f(x) = \begin{cases} -1, & -\pi < x < {}^{\pi}/_{2} \\ 0, & -{}^{\pi}/_{2} < x < {}^{\pi}/_{2}. \\ 1, & {}^{\pi}/_{2} < x < \pi \end{cases}$ (15)

(b) Expand the function $f(x) = kx,\ \ -{}^{p}/_{2} < x < {}^{p}/_{2}$ in complex form of Fourier series. (14)

(c) Write the statement of Parseval's identity of Fourier series. (06)

7. (a) The steady state temperature distribution in a thin, flat, rectangular plate of boundaries $x=0$, $x=a$, $y=0$ and $y=b$ is modeled by the differential equation (25)

$$\frac{\delta^2 u}{\delta x^2} + \frac{\delta^2 u}{\delta y^2} = 0;\ 0 < x < a,\ 0 < y < b.$$

Determine the temperature $u(x,y)$ subject to the boundary conditions:

$$u(0,y) = u(a,y) = 0;\ \ 0 < y < b\ \text{ and } u(x,0) = f(x),\ u(x,a) = g(x);\ 0 < x < a$$

(b) Using the convolution theorem evaluate $L^{-1}\left\{\frac{s}{(s^2+a^2)^2}\right\}$. (10)

8. (a) Define Laplace transform. Find the Laplace transform of *tcos2t*. (10)

(b) Find the Laplace transform of the periodic saw-tooth wave form shown in figure below: (13)

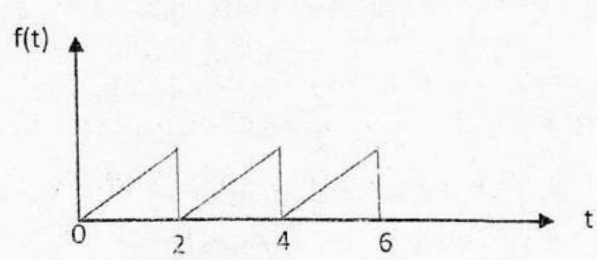

Figure 8(b)

(c) Solve the initial value problem using the Laplace Transform: (12)

$$\frac{d^2y}{dx^2} + \frac{dy}{dx} + y = e^{-t},\ y(0) = -1,\ \ y'(0) = 1\ .$$